Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti |
AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

AI platform engineer who **built AgentProbe end-to-end** - the exact systems Harvey's AI Platform team builds: multi-provider model routing (Anthropic, OpenAI, Groq, Ollama), context and session state management across multi-step agent runs, a shared 5-metric evaluation framework with CI/CD regression gates, and a 15-endpoint FastAPI platform layer that product teams build on. Built every layer from greenfield to public open-source. 8+ years production Python; TypeScript/JavaScript at Entrata (120+ sprints, 100K+ users); applied ML on 1.6M financial records. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

Model Routing & Multi-Provider Integration | Evaluation Infrastructure (LLM-as-Judge) | Context & Session State Management | Agent Architecture (LangGraph) | Platform APIs & Shared Abstractions | Python (8+ years production) | TypeScript / JavaScript | FastAPI / REST Platform Layer | CI/CD Eval Gates (GitHub Actions) | RAG + Vector Search (ChromaDB) | Azure / AWS / Docker | Production at Scale (1M+ records)

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Applied **Python, PyTorch, and OpenAI API** to automate data transformation across **1.6M financial records** - production AI on regulated data with careful output validation.
- Built and delivered **20+ full-stack modules** (**Python, TypeScript/JavaScript, PostgreSQL**) across **120+ Agile sprints** for **100K+ monthly active users** - shared platform components reused across product teams.
- Led Docker-based containerization for 30%+ of microservices; improved production query performance by **30%**; maintained **99.5% uptime** via TDD/BDD; built **5+ internal platform tools** adopted across squads - reducing resolution time by **40%**.
- Migrated client data to AWS S3 with encrypted pipelines - **100% PHI compliance, zero security incidents** across 100K+ users.
- Enforced secure coding standards; mentored engineers through peer code reviews and documentation across all major releases.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Mentored 30+ students; designed supplementary exercises; translated complex systems concepts across skill levels.

PROJECTS

AgentProbe: AI Platform Layer - Model Routing, Eval Infrastructure, Agent Runtime OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD

Built the same platform Harvey's AI Platform team builds - the foundation layer that agent and product teams run on. **Model routing**: dynamic provider selection across Anthropic, OpenAI, Groq, Ollama based on task type, latency, and capability - the infrastructure for onboarding new models fast and routing correctly per task. **Evaluation infrastructure**: 5-metric LLM-as-judge engine (factual accuracy, hallucination, retrieval, completeness, compliance) with per-run regression tracking and CI/CD gates - shared eval tooling that any team can use to measure and improve AI quality. **Context & session state**: multi-step agent runs with state preservation across tool calls and inter-agent handoffs. **Agent architecture**: 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with defined action spaces and tool routing. **Platform layer**: 15-endpoint FastAPI backend + React observability dashboard - the shared abstraction layer product teams build on top of. Open-source.
github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR: Domain-Specific NLP Platform for Medical Records

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings

NLP platform routing natural language queries to domain-specific FHIR APIs via LLMs, BERT, and embeddings - HIPAA-compliant RBAC. **~60% retrieval time reduction.** Demonstrates building AI platforms for professional workflows in regulated domains - the same pattern Harvey applies to legal. *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: Predictive ML Pipeline - 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Feature Engineering · GitHub Pages

Full ML lifecycle: feature engineering → model selection → evaluation → deployment. **85%+ accuracy.**

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML systems, experimental design, production data pipelines

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

AI Platform: LangChain, LangGraph, Multi-Provider Model Routing (Anthropic, OpenAI, Groq, Ollama), Context Management, LLM-as-Judge Evaluation, CI/CD Eval Gates, RAG Pipelines, ChromaDB, Prompt Engineering, Tool Call Architecture

Backend & Infra: FastAPI, Flask, React, PostgreSQL, MySQL, Docker, AWS (S3, EC2, RDS, Glue), Azure, CI/CD (GitHub Actions), RabbitMQ, Microservices

ML & Data: PyTorch, scikit-learn, BERT, spaCy, Hugging Face, NumPy, Pandas, Feature Engineering, Statistical Modeling

Process: Open-Source Development, TDD/BDD, Integration Testing, Git, JIRA, Agile/Scrum